

Practice 3: OVB and Technological Change

25117 - Econometrics

Universitat Pompeu Fabra

October 28th, 2024

Omitted Variable Bias

Omitted Variable Bias refers to the bias introduced in regression analysis when a relevant variable is left out of the model. This bias can distort the estimated relationships between variables and lead to incorrect conclusions. It can be represented mathematically as follows:

$$\hat{\beta}_{\text{biased}} = \beta_{\text{true}} + \frac{\text{Cov}(X, Z)}{\text{Var}(X)}$$

where:

- $\hat{\beta}_{\text{biased}}$ is the biased estimator.
- β_{true} is the true parameter of interest.
- X is the included variable.
- Z is the omitted variable.
- $\text{Cov}(X, Z)$ is the covariance between X and Z .
- $\text{Var}(X)$ is the variance of X .

Introduction to Omitted Variable Bias

Question

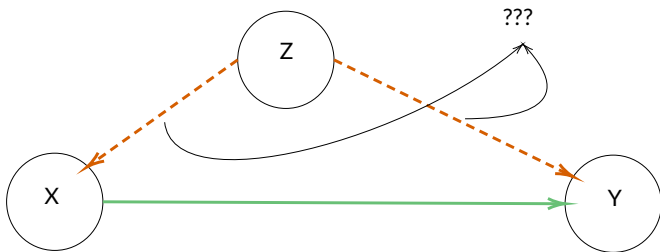
How can omitting important variables affect our analysis?

- Omitted Variable Bias occurs when a relevant variable is not included in a regression model, and this omission can lead to incorrect conclusions.
- For example, if we're studying wage changes and we omit the variable "years of education," we might mistakenly attribute wage changes to other factors, resulting in a biased estimate of the impact of omitted variables.

Omitted Variable Bias (OVB)

Question

What are the two conditions required on Z for omitted variable bias to occur?



Control Variables in Multiple Regression Analysis

Question

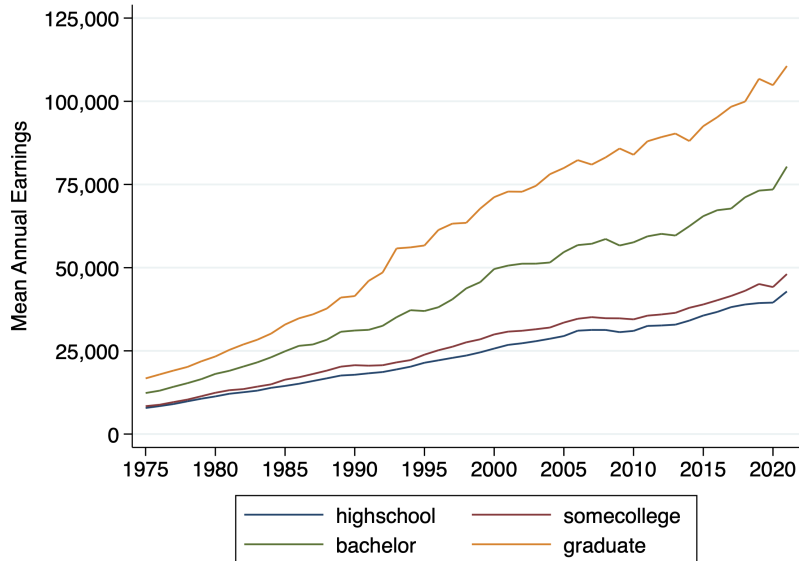
Why is it important to control for variables that might impact the dependent variable? What do we intend to achieve in these cases?

- It's crucial to control for variables that might impact the dependent variable because doing so helps us isolate the effect of our main independent variable.
- For example, when studying wages, we might control for education, experience, and industry to ensure that we're attributing changes in wages specifically to computer use and not to other factors.

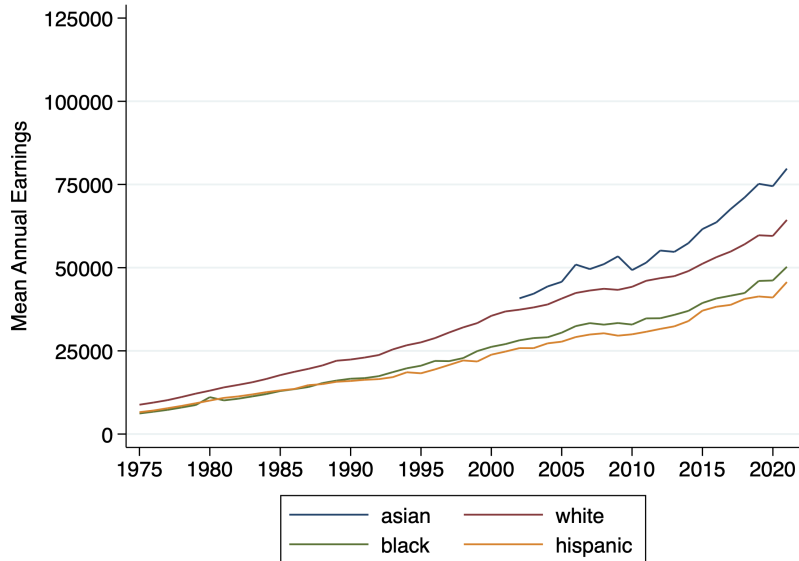
Question

What are "Control Variables"?

In Practice (1): The Returns to Education Puzzle



In Practice (1): The Returns to Education Puzzle



In Practice (1): Have Computers Changed the Wage Structure?

- Either
 - ① increased *international competition* has hurt the position of less-educated workers (less bargaining power) ; and/or
 - ② *skill-biased technological change* has affected the relative productivity highly qualified workers
- This is the 80s and it's been 10 years already since the 30-year-old Bill Gates created Microsoft. "Computers" is no longer a word meant to designate a person.
- The government charges you to understand the impact of computers on the wage structure

Questions

- What would you expect from the theory?
- What would be the gold standard experiment that would allow you to answer this question?
- How would you proceed?
- In practice, what challenges would you face?

In Practice (1): LHS vs RHS

- The previous graphs show the earnings advantage of college graduates and expanding wage differentials based on race. These stylized facts provide insights into the changing wage structure during that period.
- “Wage Structure” refers to the distribution of wages in an economy and the patterns of wage differentials among different groups of workers. It includes factors such as the earnings gap between high and low-skilled workers or wage disparities based on education or demographics.
- The computer revolution can impact wage structure in various ways. For instance, it can lead to higher wages for workers who can effectively use computers, but it may also lead to wage compression for others if they cannot adapt to the technological changes.
- The “Computer Revolution” refers to the rapid and transformative changes in the use of computers and technology in various industries and workplaces. It includes the widespread adoption of computers and their impact on work processes and productivity.

In Practice (1): Current Population Surveys (CPS)

- **Definition:** The Current Population Surveys (CPS) are a series of monthly surveys conducted by the U.S. Census Bureau.
- **Purpose:** These surveys are designed to collect a wide range of data on labor force characteristics, economic conditions, and social demographics of the U.S. population.
- **Frequency:** The CPS is conducted on a monthly basis, making it a valuable source of up-to-date information.
- **Sample Size:** The CPS includes a large and diverse sample of households and individuals, making it a representative source of data.
- **Key Features:**
 - Provides information on employment, unemployment, and wages.
 - Collects data on educational attainment, industry, occupation, and more.
 - Contains supplemental questions on various topics, including computer use in the workplace.
- **Importance:** The CPS is a critical data source as it helps monitor labor market trends and understand the socio-economic landscape of the United States.

In Practice (1): Computer Use in the 80s

Group	1984	1989
All workers	24.6	37.4
<u>Gender</u>		
Men	21.2	32.3
Women	29.0	43.4
<u>Education</u>		
Less than high school	5.0	7.8
High school	19.3	29.3
Some college	30.6	45.3
College	41.6	58.2
Postcollege	42.8	59.7

In Practice (1): Computer Use in the 80s

<u>Race</u>		
White	25.3	38.5
Black	19.4	27.7
<u>Age</u>		
Age 18–24	19.7	29.4
Age 25–39	29.2	41.5
Age 40–54	23.6	39.1
Age 55–65	16.9	26.3
<u>Occupation</u>		
Blue-collar	7.1	11.6
White-collar	33.0	48.4

In Practice (1): Computer Use in the 80s

<u>Union status</u>		
Union member	20.2	32.5
Nonunion	28.0	41.1
<u>Hours</u>		
Part-time	23.7	36.3
Full-time	28.9	42.7
<u>Region</u>		
Northeast	25.5	38.0
Midwest	23.4	36.0
South	23.2	36.5
West	27.0	39.9

In Practice (1): Empirical Strategy

Krueger (1993)'s empirical question is:

Do employees who use computers at work receive a higher wage as a result of their computer skills?

His main specification reads

$$\ln(W_i) = \beta_0 + \beta_1 C_i + \sum_{j=2}^k \beta_j X_{ji} + u_i \quad (1)$$

where

- W_i is the hourly wage of the i th individual
- C_i is a dummy variable = 1 if the i th individual uses a computer at work, and 0 otherwise
- X_{ji} intends to capture other characteristics of individual i (e.g., educational attainment, etc.)

Question

How do you read this equation? Why include a vector of variables **X**? Should standard errors be robust to heteroskedasticity? Why?

In Practice (1): Main Specification (1)

Independent variable	△△△					
	October 1984			October 1989		
	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	1.937 (0.005)	0.750 (0.023)	0.928 (0.026)	2.086 (0.006)	0.905 (0.024)	1.094 (0.026)
Uses computer at work (1 = yes)	0.276 (0.010)	0.170 (0.008)	0.140 (0.008)	0.325 (0.009)	0.188 (0.008)	0.162 (0.008)
Years of education	—	0.069 (0.001)	0.048 (0.002)	—	0.075 (0.002)	0.055 (0.002)
Experience	—	0.027 (0.001)	0.025 (0.001)	—	0.027 (0.001)	0.025 (0.001)
Experience-squared ÷ 100	—	-0.041	-0.040	—	-0.041	-0.040
▽▽▽						

In Practice (1): Empirical Strategy Cont'd

Krueger (1993) is worried that ability might confound the main effect:

A critical concern in interpreting the OLS regressions reported above is that workers who use computers on the job may be abler workers [...]. The finding that the computer wage differential is attenuated when covariates are included in the OLS regressions suggests that important variables may be omitted that are positively correlated with both computer use and earnings.

To capture computer ability, the author estimates the following equation

$$\ln(W_i) = \beta_0 + \beta_1 C_{w,i} + \beta_2 C_{h,i} + \beta_3 C_{hw,i} + \sum_{j=4}^k \beta_j X_{j,i} + u_i \quad (2)$$

where

- W_i is the hourly wage of the i th individual
- $C_{w,i}$ is a dummy variable = 1 if the i th individual uses a computer at work
- $C_{h,i}$ is a dummy variable = 1 if the i th individual uses a computer at home
- $C_{hw,i}$ is a dummy variable = 1 if the i th individual uses a computer at home and at work
- X_{ji} intends to capture other characteristics of individual i (e.g., educational attainment, etc.)

In Practice (1): Controlling for a computer-ability proxy

Type of computer use	October 1984 (1)	October 1989 (2)	Percent of sample, 1989 (3)
Uses computer at work	0.165 (0.009)	0.177 (0.009)	39.8
Uses computer at home	0.056 (0.021)	0.070 (0.019)	12.5
Uses computer at home and work	0.006 (0.029)	0.017 (0.023)	8.6
Sample size	13,335	13,379	

Question

Interpret these coefficients and their associated standard errors. What is the omitted category (comparison group) here?

In Practice (2): Have Pencils changed the Wage Structure Too?

- The study by Krueger (1993) examines the relationship between computer use on the job and wage differentials.
- Krueger's findings suggest that computer users earn 15 to 20 percent more than nonusers after controlling for standard worker attributes.
- The study also attempts to address whether this wage differential primarily reflects a "causal" relationship between computer use and wages or is a result of unobserved worker heterogeneity.
- Di Nardo and Pischke (1997) replicates Krueger's analysis using data from German workers.
- The results in Germany are similar to those found by Krueger in the United States.
- The computer wage premium in Germany is estimated to be around 17 percent in 1991.

In Practice (2): Have Pencils changed the Wage Structure Too?

- In Di Nardo and Pischke (1997), regression models are used to analyze the wage differentials associated with other tools such as pencils, calculators, or telephones (“white-collar” tools) vs. hammers, screwdrivers (“blue-collar” tools)
- The regression model replicates Krueger (1993) for different tools:

$$\ln(W_i) = \beta_0 + \beta_1 Tool_i + \sum_{j=2}^k \beta_j X_{ji} + u_i \quad (3)$$

- After conditioning on the set of covariates $\mathbf{X}$, if computers yield a higher wage as a result of computer skills, we should also observe that the use of pencils yields negligible wage differentials (everyone knows how to use a pen).
- Is it about return to skills or selection effects?

In Practice (2): Effects of each tool

Independent variable	Germany 1979	Germany 1985–86	Germany 1991–92	Germany 1979	Germany 1979	Germany 1985–1986	Germany 1991–1992
Occupation indicators	No	No	No	501	501	742	1071
Grades and father's Occupation ^a	No	No	No	No	Yes	No	No

Tools entered separately

Computer	0.112 (0.010)	0.157 (0.007)	0.171 (0.006)	0.025 (0.011)	0.022 (0.011)	0.076 (0.008)	0.083 (0.007)
Calculator	0.087 (0.007)	0.128 (0.006)	0.129 (0.006)	0.027 (0.008)	0.025 (0.008)	0.061 (0.007)	0.054 (0.006)
Telephone	0.131 (0.006)	0.114 (0.006)	0.136 (0.006)	0.060 (0.007)	0.057 (0.007)	0.059 (0.007)	0.072 (0.007)
Pen/pencil	0.123 (0.006)	0.112 (0.006)	0.127 (0.006)	0.055 (0.007)	0.052 (0.007)	0.055 (0.007)	0.050 (0.007)
Work while sitting	0.106 (0.006)	0.101 (0.007)	—	0.042 (0.008)	0.041 (0.008)	0.036 (0.008)	—
Hand tool (e.g., hammer)	-0.117 (0.007)	-0.086 (0.006)	-0.091 (0.006)	-0.048 (0.009)	-0.045 (0.009)	-0.020 (0.008)	-0.020 (0.008)

In Practice (2): Exploring the results

- To account for potential selection effects related to occupations, the analysis includes occupation dummy variables.
 - These occupation dummies help control for variations in wage structures across different jobs.
 - Even within narrowly defined occupations, significant wage differentials for office tools persist.
 - Ability controls, such as grades and academic achievements, are included in the analysis to address potential selection effects.
 - Results suggest that ability controls are not proxies for the skills relevant in the workplace.
 - This indicates that the estimated wage differentials for tools, even within narrowly defined occupations, are not solely due to the workers' academic abilities.
- Strong selection effects seem to persist.